

Question 1

2 marks

101

A pizza with a diameter of 15 inches is cut into equal slices, each with a central angle of 36° .
Determine the length of the crust on the outer edge of one slice of pizza.

Question 2

a) 1 mark b) 2 marks

102
103

There are 9 girls and 7 boys in a math class from which a committee of 5 is to be chosen.

- a) How many different committees of 5 can be formed if one of the boys, William, must be on the committee?

- b) How many different committees of 5 can be formed if there must be 2 girls and 3 boys on the committee?

Question 3

3 marks

104

Solve the following equation over the interval $[0, 2\pi]$:

$$\sin^2 \theta + 6 \sin \theta - 2 = 0$$

Question 4

4 marks

105

Solve:

$$6(5)^{3x+2} = 9^{2-x}$$

Note: A calculator is not required for the remaining test questions.

Question 5

4 marks

106

Solve $(2 \sin \theta - 1)(\sin \theta + 1) = 0$ where $\theta \in \mathbb{R}$.

Question 6

1 mark

107

The roots of the polynomial equation $3(x - 2)(x + 1)^2 = 0$ are $x = 2$ and $x = -1$.

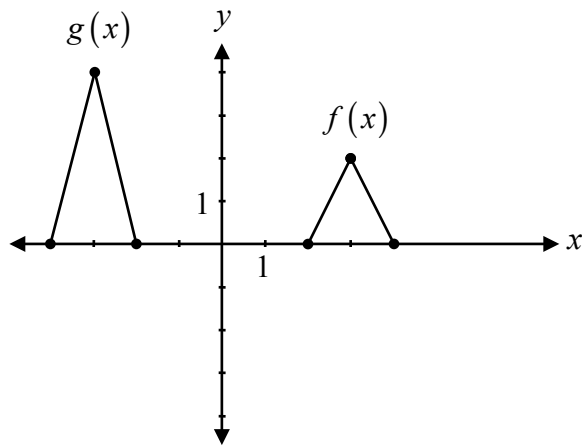
Explain what these roots represent on the graph of $p(x) = 3(x - 2)(x + 1)^2$.

Question 7

2 marks

108

Determine an equation for $g(x)$ as a transformation of $f(x)$.



$g(x) =$ _____

Question 8

1 mark

109

A student must determine the factors of $5x^4 - 2x^3 + 4x - 1$. He used 5, -2 , 4, and -1 as the coefficients of the polynomial when using synthetic division.

Explain the student's error.

Question 9

2 marks

110

Describe the transformations of $y = f(x)$ when asked to sketch the graph of $y = -f(x - 4)$.

Question 10

3 marks

111

Prove the identity below for all permissible values of θ :

$$\sin \theta + \frac{\cos \theta}{\tan \theta} = \frac{1}{\cos \theta \tan \theta}$$

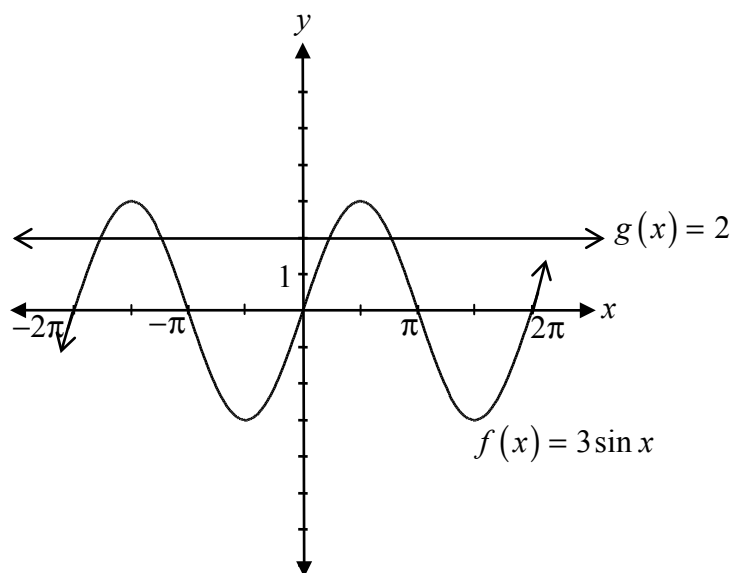
Left-Hand Side	Right-Hand Side

Question 11

1 mark

112

Describe how to use the graphs of $f(x) = 3 \sin x$ and $g(x) = 2$ to solve the equation $3 \sin x = 2$.



Question 12

1 mark

113

A hockey arena has 5 doors.

Determine the number of ways that you can enter through one door but exit through a different door.

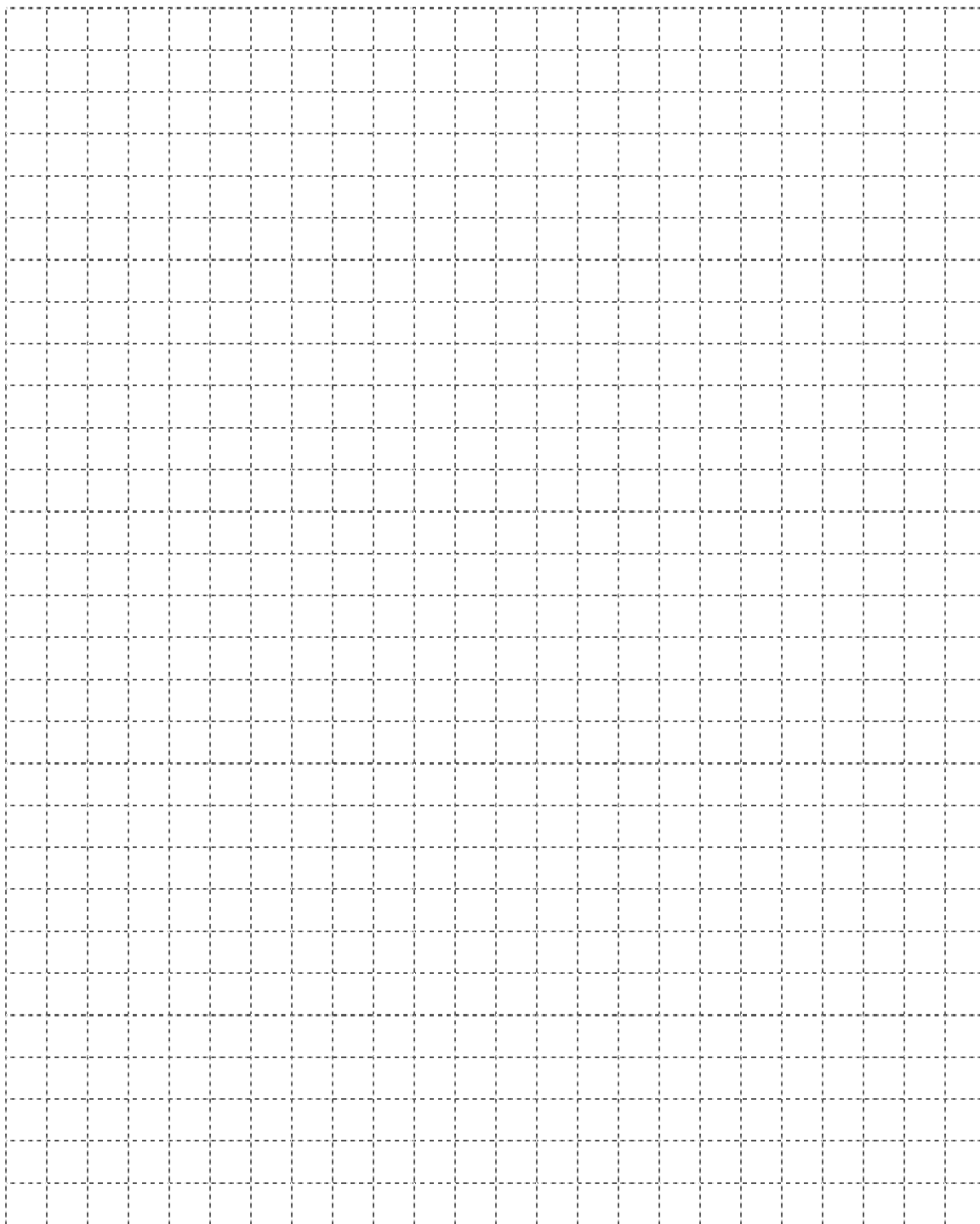
Question 13

2 marks

114

Given that $(x + 3)$ is one of the factors, express $2x^3 + 7x^2 + 2x - 3$ as a product of factors.

No marks will be awarded for work done on this page.



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